

THE EVAL GAP · A THREE-PART SERIES

The Eval Gap

Three posts, one argument

01

Metrics, Metrics, Metrics

Generic metrics give the feeling of measurement without the reality.

The field to read is psychometrics.

Running a test is not measuring

01

Metrics are the output, not the stack

WE'VE SEEN THIS MOVIE



The comfort of a number

Psychometrics has done this for a century

The guessing floor

Same raw score, different reality

When the construct is absent

Not academic — becoming standard practice

02



A floor for defensible practice



Construct vs. criterion

A gate with no validity



Protocol design

Six ways a score lies

The model knows it's being tested

READ IT AS A GAP ANALYSIS

'Emerging practice' = not yet adopted

ONE VOCABULARY, FIVE LAYERS

Transcript $\subset$ trajectory $\subset$ rollout $\subset$ logs/traces

Earn the claim before you make it

In NIST AI 800-2, where do most benchmark-evaluation failures happen?

- A Stage 1 — defining what you're actually measuring
- B Stage 2 — running the evaluation harness
- C Stage 3 — reporting the results
- D In the model weights themselves

Models A and B score 73% vs 68% raw. B has a higher guessing floor (0.35 vs 0.04) and beats A on hard items. What does that tell you?

- A** B is the stronger reasoner — it wins on the hard questions
- B** Most of B's hard-item correct answers are indistinguishable from hallucinations
- C** A and B are interchangeable for deployment
- D** B was simply tested on easier items

03

Two accuracies, two error bars

Why $\sqrt{n \cdot t}$ is the wrong denominator

A Rasch model for benchmarks

DIAGNOSTICS, NOT JUST BARS

What the GLMM surfaces

What goes in the validation report



The work the field skips